

■ B.6.2 The Main Loop

- Read in input.
- Make translations specified by aliases.
- Apply `$Pre` function, if defined.
- Assign `In[n]`.
- Evaluate expression.
- Apply `$Post` function, if defined.
- Assign `Out[n]`.
- Apply `$PrePrint` function, if defined.
- Print expression, if it is not `Null`.
- Increment `$Line`.

An important point is that when the value of `Out[n]` (and thus `%`) is assigned, certain formatting functions are stripped off. In addition to the format types `OutputForm`, `InputForm`, `CForm`, etc., the following formatting functions are removed: `Short`, `TreeForm`, `MatrixForm`, `TableForm`. As a result, even if you type `Short[expr]`, the value assigned to `Out[n]` will still be `expr`.

If your input consists of a compound expression ending with `;`, then `Null` will be returned, and no output will be printed. The value of `Out[n]` will nevertheless be assigned to be the last element in the compound expression.